Sajini Tharushika

Information Technology Undergraduate

Email: sajinitharushika3655@gmail.com | **Phone:** +94 77 653 76 27

LinkedIn: LinkedIn | **GitHub:** GitHub | **Portfolio:** Portfolio

Profile

Information Technology undergraduate at the University of Moratuwa seeking a Software Engineering internship. Hands-on project experience building full-stack applications with Java, Spring Boot, React, and PostgreSQL. Implemented REST APIs, role-based authentication, asynchronous job processing, and automated tests across individual and team projects. Strong foundations in object-oriented programming, data structures, and algorithms.

Education

University of Moratuwa, Sri Lanka 2023 – 2027

B.Sc. (Hons) in Information Technology **Current GPA: 3.4**

Sujatha Vidyalaya, Matara G.C.E. Advanced Level – Mathematics Stream

Z-Score: 1.87

Skills

- **Programming:** Java, Python, JavaScript, C++, TypeScript
- **Web & Backend:** Spring Boot, Spring Security, React, Next.js, Node.js, PHP, REST APIs, WebSockets, STOMP, HTML, CSS, Tailwind CSS
- **Databases & ORM:** PostgreSQL, MySQL, MongoDB, Redis, SQL, Prisma, JPA/Hibernate
- **Software Engineering:** Object-Oriented Programming, Data Structures & Algorithms, Software Testing, Vitest, Playwright, Authentication, Agile/Scrum
- **Distributed Systems & Infrastructure:** RabbitMQ, Docker, Kubernetes, GitHub Actions, CI/CD, Asynchronous Processing, Observability, Prometheus, Grafana, WSO2 API Manager
- **Tools & Development:** Git, GitHub, VS Code, Vercel, Render
- **AI & Data:** Pandas, NumPy, Scikit-learn, Machine Learning Fundamentals, AI API Integration, FastAPI, Groq API
- **Additional Technologies:** OpenGL, GLSL, WebGPU, WGSL, Arduino, ESP32, Embedded Systems

Projects

Nexora – AI-Enhanced Project Management System Group Project

- Co-developed a role-based project management application for managing projects, tasks, tickets, and communication between administrators, managers, developers, and clients.
- **My Contribution:** Built the administrator dashboard, user registration and role assignment, and manager-side project management. Implemented an AI chat interface for conversation summaries and summary-to-ticket creation.
- **Tech:** Java, Spring Boot, Spring Security, React, PostgreSQL, Python, FastAPI, Groq API, RabbitMQ, WebSockets/STOMP, JWT, Docker
  GitHub: `github.com/Sajini3655/Nexora`
  Live Demo: `nexora-three-peach.vercel.app/login`

FlowForge – API Management & Distributed Workflow Platform Individual Project

- Built a Spring Boot platform integrating API management with asynchronous job processing through WSO2 API Manager and RabbitMQ.
- Implemented transactional outbox publishing, idempotent requests, retry queues, dead-letter handling, and Redis-based locking to handle failures and coordinate workers.
- Added JWT authentication, Prometheus/Grafana monitoring, and containerized deployment configurations using Docker and Kubernetes.
- **Tech:** Java, Spring Boot, PostgreSQL, WSO2 API Manager, RabbitMQ, Redis, Docker, Kubernetes, Prometheus, Grafana, GitHub Actions
  GitHub: github.com/Sajini3655/Flowforge

PhysicsLab – Bilingual A/L Physics Learning Platform Group Project

- Co-developed an English/Sinhala A/L Physics learning platform with interactive practicals, assessments, and progress tracking.
- **My Contribution:** Implemented interactive practicals, physics calculation checks, and browser-local persistence, with automated verification using Vitest and Playwright.
- **Tech:** Next.js, React, TypeScript, Tailwind CSS, Vitest, Playwright
  GitHub: `github.com/vishiiboy/physicslab`
  Live Demo: `physicslab-eta.vercel.app/en`

FabricLab – Interactive Material & Physics Laboratory Individual Project

- Built an interactive cloth simulation with TypeScript and WebGPU compute shaders, implementing an XPBD solver, collision handling, 3D interaction, and dynamic pinning.
- Added configurable material presets, deformation visualization, saved experiments, and a 5K–100K-particle benchmark suite with frame-time statistics and CSV export.
- **Tech:** TypeScript, WebGPU, WGSL, XPBD, GPU Compute Shaders, WebGL/Graphics Mathematics, Webpack, Node.js Test Runner, GitHub Actions, GitHub Pages
  GitHub: `github.com/Sajini3655/fabriclab`
  Live Demo: `sajini3655.github.io/fabriclab/`

AURA – Full-Stack Fashion E-Commerce Web Application Individual Project

- Built a full-stack fashion e-commerce application with product browsing, cart management, checkout, rewards, and voucher redemption.
- Implemented API routes, Prisma/MongoDB database operations, JWT authentication with bcrypt password hashing, and a spin-the-wheel rewards feature.
- **Tech:** Next.js 15, React, JavaScript, Tailwind CSS, Prisma, MongoDB, JWT, bcrypt
  GitHub: github.com/Sajini3655/Aura

LaserTunes – Laser-Based Musical Instrument Group Project

- Co-developed a 12-laser musical instrument with instrument switching, percussion effects, and loop recording using Arduino, ESP32, sensors, and audio hardware. Presented at **FIT EXPO 2025**.
- **Tech:** Arduino Mega, ESP32, C++, VS1053B, Piezo Sensors, Embedded Systems
  GitHub: github.com/Sajini3655/LaserTunesCode

Solar – Computer Graphics Animation Project Group Project

- Co-developed a 3D animation using C/C++ and OpenGL, incorporating GLSL shaders, textures, and visual effects.
- **Tech:** C, C++, GLSL, OpenGL
  GitHub: github.com/Sajini3655/solar

Fitness First – Gym Website Individual Project

- Built a fitness website with trainer details, services, meal plans, pricing, user accounts, and an interactive BMI calculator.
- Implemented PHP/MySQL registration and login, database-backed account functionality, and asynchronous requests using JavaScript fetch().
- **Tech:** HTML, CSS, JavaScript, PHP, MySQL
  GitHub: github.com/Sajini3655/fitnessfirst
  Live Demo: `sajini3655.github.io/fitnessfirst/`

Leadership & Extracurricular Activities

- **Editorial Head** – IEEE Computer Society
- **Web & Technology Pillar Member** – MoraSpirit
- **Editorial Board Member** – Leo Club of University of Moratuwa
- **Event Coordinator** – Leo Club of University of Moratuwa
- **Editorial Volunteer** – Rotaract Club

Referees

Mr. Chirath Perera — Senior Software Engineer, Creative Software
Phone: +94 76 576 3993

Ms. Nipuni Premadasa — Lecturer, University of Moratuwa
Phone: +94 70 170 3862 Email: nipunip@uom.lk